📊 Executive Performance Report: Global Sales & Profitability

Lead Analyst: Abdullah F.

Tech Stack: Power BI (Desktop & Service), DAX, Power Query, Star Schema Modeling

---

📘 1. Business Problem & Objectives

The objective of this analysis was to evaluate the **Year-to-Date (YTD) performance** against the **Previous Year-to-Date (PYTD)** to identify growth stagnation and profitability leaks. The business required a granular view of regional performance and customer concentration to determine if revenue growth was driven by sustainable market expansion or high-risk account reliance.

🏗 2. Data Architecture & Modeling

I implemented a **Star Schema** to optimize query performance and ensure filter integrity across all visuals.

- **Fact Table:** `Fact_Sales` (Contains transactional data, quantities, and gross profit metrics).
- **Dimension Tables:**
  - `Dim_Date`: A custom DAX calendar table supporting Time Intelligence.
  - `Dim_Geography`: Regional mapping for US, UK, and Ireland.
  - `Dim_Accounts`: Customer-level attributes.
- **Key Modeling Logic:** Established a 1:N relationship between Dimensions and Facts to prevent data ambiguity.

□ 3. Advanced DAX Gallery

To power the Waterfall and Scatter plots, I developed custom measures to calculate variance and margin health:

YTD vs PYTD Variance:

$$Variance = [Total Sales YTD] - [Total Sales PYTD]$$

Gross Profit Margin %:

$$GP\% = DIVIDE([Total Gross Profit], [Total Sales], 0)$$

Note: These measures allow the dashboard to dynamically update based on the "Year" slicer (currently set to 2022) without manual formula adjustments.

📈 4. Strategic Deep Dive

Geographic Performance (The "Long Tail" Effect)

While the **United States ($35.6k)** is the primary revenue engine, the **Bottom 10 YTD vs PYTD** treemap reveals critical "At-Risk" zones. Countries like **Ghana** and **Spain** show the highest negative variance, indicating a loss in market share compared to the previous period.

Profitability vs. Volume (Scatter Plot Analysis)

The **Value YTD and GP% by Account** scatter plot reveals that the majority of revenue is generated at a **40% GP margin**. However, outliers in the lower-right quadrant represent high-volume accounts with sub-par margins—these are prime candidates for price optimization.

Customer Concentration Risk

The analysis confirms a **Low Concentration Risk**. With the top 5 customers contributing roughly **$300 each** against a **$35k+ market**, the business is highly resilient to individual churn.

🚀 5. Recommendations

1. **US Market Retention:** Given that the US drives 80% of revenue, implement a loyalty program for the $250–$350 spend tier to secure the "core" volume.
2. **Marginal Recovery:** Conduct a cost-of-goods-sold (COGS) audit for accounts falling below the **0.35 GP% threshold** identified in the scatter plot.
3. **Regional Pivot:** Investigate the supply chain or marketing spend in **Ghana and Cambodia**, as they are currently the largest detractors from YTD growth.